

A Variational Quantum Algorithm for Ordered SVD

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Abstract—Singular value decomposition (SVD) is fundamentally important and broadly useful in both quantum and classical computing. There are several quantum algorithms known for SVD. Variational quantum algorithms with parametric quantum circuits (PQC) are the most promising approach to find SVD with near-term quantum computers. This paper reports a new method for quantum SVD (QSVD) that finds the singular vectors ordered by the magnitude of singular values by adding extra CNOT gates and a new local cost function design. This ordering is important because the singular vectors with the larger singular values is more representative for the given information. The ordering property of this approach is investigated with the standard Iris dataset by numerical simulations of 4-qubit states. Methods for choosing appropriate choices for the cost function hyperparameter and cost function terms as well as an application example of a quantum encoder are also discussed.

Index Terms—singular value decomposition, variational quantum algorithm, parametric quantum circuit

I. INTRODUCTION

Quantum data processing is an essential function of various quantum algorithms that manipulate complex data, which includes principal component analysis [1] and independent component analysis [2]. Of these methods, singular value decomposition (SVD) is one of the most important tools for both classical and quantum computing. In classical computing, it has been used for various applications including image processing, principal component analysis (PCA), and recommendation systems. For quantum information, SVD can be used to calculate the Schmidt decomposition of a quantum state. This allows analysis of the entanglement of the state and calculation of the von Neumann entropy. It can be also used to approximately encode a quantum state into lower dimensions. Considering the usefulness of SVD, it is important to find an efficient quantum algorithm to calculate it.

There are physical limits in running algorithms on a quantum computer. Currently, only noisy intermediate scale quantum (NISQ) machines are available, as explained in [3]. These machines do not consist of fault-tolerant qubits so

that the computation is limited to short gate depth circuits due to the rapid decoherence. This prohibits NISQ machines from running more quantum resource-intensive algorithms for calculating SVD, such as the ones proposed in [4] [5].

One of the major approaches for using NISQ devices is variational quantum algorithms (VQAs). VQA is a type of hybrid algorithm, where quantum computing based on a PQC is used to calculate a cost function. This cost function and the parameters of the circuit are handled by a classical computer to iteratively optimize the cost function. Since the quantum hardware is only used to calculate the cost function, VQAs may require only shallow depth circuits. The circuit and cost function are designed so that when the cost function is minimized, the resulting PQC with optimal parameters performs a certain task or calculates a meaningful solution.

There have been two major variational quantum algorithms proposed for calculating SVD. The quantum singular value decomposer in [6] used a quantum state on a set of qubits divided into two registers. Two PQCs, each acting on one of the registers, are trained using a local cost function so that the measurement of each quantum register yielded the same results. This allowed calculation of the right and left singular vectors up to a global phase in the form of the trained PQCs. The singular values are calculated from the measurement probabilities of the trained circuit.

The variational quantum algorithm for singular value decomposition (VQSVD) [7] used a global cost function that exploits the rearrangement inequality and the Ky Fan theorem to calculate the SVD of a matrix. This approach allows the singular vectors to be known in order of the singular value magnitude without extra measurements after training.

This paper proposes an enhanced VQA that calculates SVD with the singular vectors ordered with respect to the corresponding singular values. The circuit design of QSVD is changed so that only one of the two registers are measured to calculate the cost function originally proposed. This opens up the other register to be measured and used to force the desired ordering. A new local cost function term is added

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to the original QSVD cost function. The local cost function prevents barren plateaus in shallow quantum neural networks [8], allowing the circuits to be trained. Our algorithm effectively combines some characteristics of QSVD and VQSVD. To show the effects of the new cost function term, numerical simulations of SVD of random 4-qubit states and PCA of the Iris dataset was conducted.

II. ORDERED QSVD

A. The QSVD Algorithm

Mathematically, the intuition behind the QSVD algorithm is as follows: Consider a quantum state of a bipartite system $|\psi\rangle_{AB}$. Let the Hilbert space of subsystem A be $\mathcal{H}_A$ and subsystem B be $\mathcal{H}_B$. One can express $|\psi\rangle_{AB}$ using the computational bases $\{|j\rangle_A\}$ and $\{|k\rangle_B\}$ of $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively:

$$|\psi\rangle_{AB} = \sum_{j,k} a_{jk} |j\rangle_A |k\rangle_B \quad (1)$$

Here, $a_{jk} = \langle j|_A \langle k|_B |\psi\rangle_{AB}$, $\sum_{j,k} |a_{jk}|^2 = 1$. The probability amplitudes a_{jk} can be thought of as elements of a matrix M , M_{jk} . Then, there exists an SVD of M :

$$M = U \Sigma V^\dagger \quad (2)$$

Σ is a diagonal matrix with positive real entries. U and V are unitary matrices. The $\dagger$ is the conjugate transpose. Then (1) can be written using the matrix components of U , Σ , V as:

$$\begin{aligned} |\psi\rangle_{AB} &= \sum_{j,k} a_{jk} |j\rangle_A |k\rangle_B \\ &= \sum_i \sigma_{ii} \left(\sum_j u_{ji} |j\rangle_A \right) \left(\sum_k v_{ki}^* |k\rangle_B \right) \\ &= \sum_i \sigma_{ii} |u_i\rangle_A |v_i^*\rangle_B \end{aligned} \quad (3)$$

Here, $|u_i\rangle$, $|v_i\rangle$ are the column vectors of U and V , and $|v_i^*\rangle$ is the complex conjugate of $|v_i\rangle$. Therefore, $\{|u_j\rangle\}$, $\{|v_k\rangle\}$ are sets of orthonormal vectors in the Hilbert spaces of subsystem A and B respectively. They are also the left and right singular vectors of M . The diagonal entries of Σ , σ_{ii} , are the singular values of M . Then, there exist basis transforms in each Hilbert space that transforms from $\{|u_i\rangle\}$, $\{|v_i^*\rangle\}$ to the computational basis of each Hilbert space up to some phase term. For instance,

$$\sum_j e^{i\alpha_j} |j\rangle_A \langle u_j|, \sum_k e^{i\beta_k} |k\rangle_B \langle v_k^*| \quad (4)$$

To find such transforms, the QSVD algorithm uses a circuit initialized with $|\psi\rangle_{AB}$ and parameterized quantum circuits on the subsystems A and B . The scaling of the circuit depending on the qubit number will depend on the architecture of the PQC.

The QSVD algorithm uses the cost function (5), which measures the Hamming distance between the computational basis state measured in the upper and lower registers. n_A and n_B are the number of qubits in each subsystem. Minimization of the cost function causes matching computational basis states

to be measured with high probability. In other words, when the cost function is minimized to 0 there should be at most $2^{\min(n_A, n_B)}$ basis states measured. This effectively finds the SVD with a cost function

$$\text{QSVD Cost Function} = \sum_{i=1}^{\min(n_A, n_B)} \frac{1 - \langle Z_i^A \otimes Z_i^B \rangle}{2}. \quad (5)$$

Here $Z_i^A(Z_i^B)$ is the Pauli Z operator on the i^{th} qubit of the $A(B)$ register. Note that this cost function is a local cost function, so it does not suffer from the barren plateau problem in $O(\log(n_A + n_B))$ depth circuits.

After the circuit is optimized, there are some extra steps needed to calculate the singular values and vectors. The optimized PQC_A and PQC_B are interpreted to be the unitary transforms in (4). Then $PQC_A^\dagger |j\rangle = e^{-i\alpha_j} |u_j\rangle$ and $PQC_B^\dagger |k\rangle = e^{-i\beta_k} |v_k^*\rangle$. The singular values are $\{\sqrt{P(|i\rangle_A |i\rangle_B)}\}_i$, the square root of the probabilities to measure computational basis state $|i\rangle_A |i\rangle_B$. In this formulation additional measurements after training is needed to find singular vectors in the order of the singular values.

B. The Ordered QSVD Algorithm

This paper suggests the ordered QSVD algorithm(OQSVD), which enhances the QSVD algorithm in two ways in calculation of SVD with the singular vectors ordered with respect to the singular values. We define 'ordered' to mean that the i^{th} largest singular value is $\sqrt{P(|i-1\rangle_A |i-1\rangle_B)}$ and the corresponding singular vector is found from $PQC_A^\dagger |i-1\rangle_A$ and $PQC_B^\dagger |i-1\rangle_B$ after training. The following discussion assumes $n_A \leq n_B$, but can be extended to when $n_A > n_B$.

The first change is to add CNOT gates to the QSVD circuit, as shown in Fig.1. This circuit was discussed in [6] as a possible way of encoding quantum states. This circuit also allows for the cost function in (5) to be calculated only using measurements of register B :

$$\begin{aligned} \langle Z_i^A \otimes Z_i^B \rangle &= \langle \phi | Z_i^A \otimes Z_i^B | \phi \rangle \\ &= \langle \phi | \text{CNOT}_{q_{A_i} \rightarrow q_{B_i}}^\dagger I_i^A \otimes Z_i^B \text{CNOT}_{q_{A_i} \rightarrow q_{B_i}} | \phi \rangle \end{aligned} \quad (6)$$

$\text{CNOT}_{q_{A_i} \rightarrow q_{B_i}}$ is a CNOT gate with the control qubit q_{A_i} and target qubit q_{B_i} . $|\phi\rangle$ is the quantum state at the end of the circuit shown in Fig. 1.

This addition of gates frees up the top register to be measured and used for another cost function term. An additional cost term is the second enhancement suggested by this paper. Note the singular values correspond to the square root of the measurement probability of each computational basis state of the A register after training the circuit. Given real numbers $\{\alpha_i\}$ satisfying $\alpha_j \geq \alpha_k, \forall j > k$, and $\{P(|i\rangle_A)\}$, $\sum_i \alpha_i P(|i\rangle_A)$ is maximized only when $P(|j\rangle_A) \geq P(|k\rangle_A), \forall j > k$. Here, $P(|i\rangle_A)$ is the probability to measure the A register as the computational basis state represented by the bit string of i in binary.

However, the term $\sum_i \alpha_i P(|i\rangle_A)$ can be a global cost function. Thus, it may suffer from barren plateau problems, not allowing the PQCs to be trained. Consider a form of

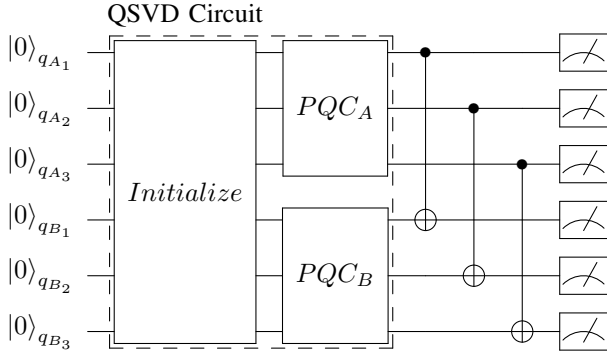


Fig. 1. The QSVD circuit and extra CNOT gates. This example has a total of 6 qubits, divided into two 3 qubit subsystems. The circuit in the dotted line is the QSVD. When the number of qubits in register A and B are different, there should be $\min(n_A, n_B)$ CNOT gates added to the QSVD circuit. These gates should have control qubits on the register with the lesser number of qubits. The summation in the QSVD term in (10) should run over the target qubits of the additional CNOT gates, and the summation in the ordering term should run over the control qubits of the additional CNOT gates.

local cost function $\sum_j \beta_j \langle Z_j^A \rangle$. β_j must satisfy the following relation for this type of local cost function to be equivalent to $\sum_i \alpha_i P(|i\rangle_A)$:

$$\alpha_i = \sum_{j=1}^{n_A} \beta_j (-1)^{b_j^i} \quad (7)$$

b_j^i is the j^{th} least significant bit of the binary representation of i . Note that the value i is represented in terms of b_j^i as

$$i = \sum_{j=1}^{n_A} 2^{j-1} \frac{1 - (-1)^{b_j^i}}{2} \quad (8)$$

Thus, if we set $\beta_j = 2^{j-1}$, then we have:

$$\alpha_i = \sum_{j=1}^{n_A} 2^{j-1} (-1)^{b_j^i} = \left(\sum_{j=0}^{n_A-1} 2^j \right) - 2i \quad (9)$$

α_i decreases linearly with i . Therefore, for $\sum_j \beta_j \langle Z_j^A \rangle$ to be maximized, $P(|j\rangle_A) \geq P(|k\rangle_A) (\forall j < k)$ needs to be satisfied. This forces the ordering of singular vectors in the order of the corresponding singular values. When optimized correctly, the singular vector with the largest singular value should be $PQCA |0\dots 0\rangle$ and $PQCB |0\dots 0\rangle$.

The cost function used in ordered QSVD is (10). This combines two terms; the QSVD term and the ordering term. In order to keep the limits of the cost function between $[-1, 1]$, some normalizing coefficients were added. Also a hyperparameter $\lambda \in [0, 1]$ is used to change the relative size of each term.

$$\text{Ordered QSVD Cost Function} = - \left\{ \frac{\lambda \sum_{i=1}^{n_A} \langle Z_i^B \rangle}{n_A} + \frac{(1 - \lambda) \sum_{j=1}^{n_A} 2^{j-1} \langle Z_j^A \rangle}{2^{n_A} - 1} \right\} \quad (10)$$

III. NUMERICAL SIMULATIONS

A. Ordered QSVD of Random 4-qubit States

To test the ordered QSVD algorithm, 100 random 4-qubit states were decomposed individually. Each state was created by sampling 32 values from a continuous uniform distribution $U(-0.5, 0.5)$. Each value became either the real or imaginary part of a probability amplitude of the state. The state was then normalized. Each 4-qubit state was divided into two 2-qubit subsystems. A 2-qubit quantum combinatorial ansatz (QCA) [9] was used for each PQC. This ansatz is a stabilizer ansatz and will span the entire Hilbert space. As shown in Fig. 2, each PQC has 6 parameters and so the total qSVD circuit has 12 parameters. To see the effect of λ on the results, λ was changed from 0 to 1 in 0.05 increments. Optimization of the cost function was done using the Powell algorithm implemented in the python scipy library. Qiskit [10] was used for the quantum circuit simulation.

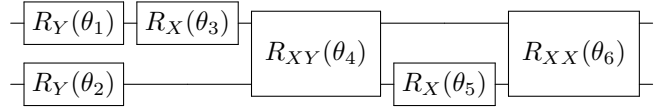


Fig. 2. The 2-qubit QCA

One example result is shown in Fig. 3. One 4-qubit state was decomposed using classical calculations, ordered QSVD with $\lambda = 1$, and ordered QSVD with $\lambda = 0.15$. The $\sqrt{P(|i-1\rangle_A)}$ values were assigned to the i^{th} singular value, σ_i . All of the singular values were ordered for $\lambda = 0.15$ while they were not ordered for $\lambda = 1$.

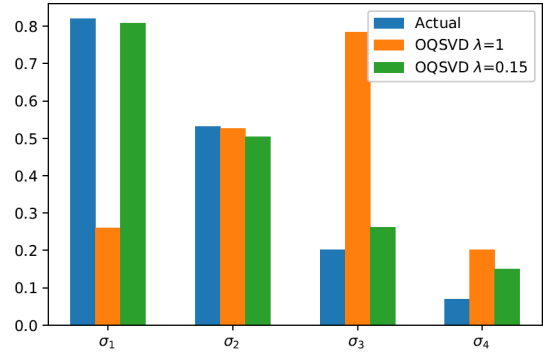


Fig. 3. Calculated Singular Values of a 4-qubit state using classical methods, QSVD (OQSVD with $\lambda = 1$), and Ordered QSVD ($\lambda = 0.15$).

The results were analyzed with two metrics. The first is the number of ordered singular vectors. The singular vectors were best ordered at $\lambda = 0, 0.05, 0.1$. When $\lambda = 1$, the cost function is equal to the QSVD term. In this case, we expect that the singular vectors are ordered randomly. This should give an average of 1 ordered singular vector, which is similar to the average value seen in the simulation.

The second metric is the QSVD term value. This value summarizes how well the trained PQCs found the singular

vectors of the matrix. The QSVD term value decreased as λ increased and had an average close to 0 at $\lambda = 0$.

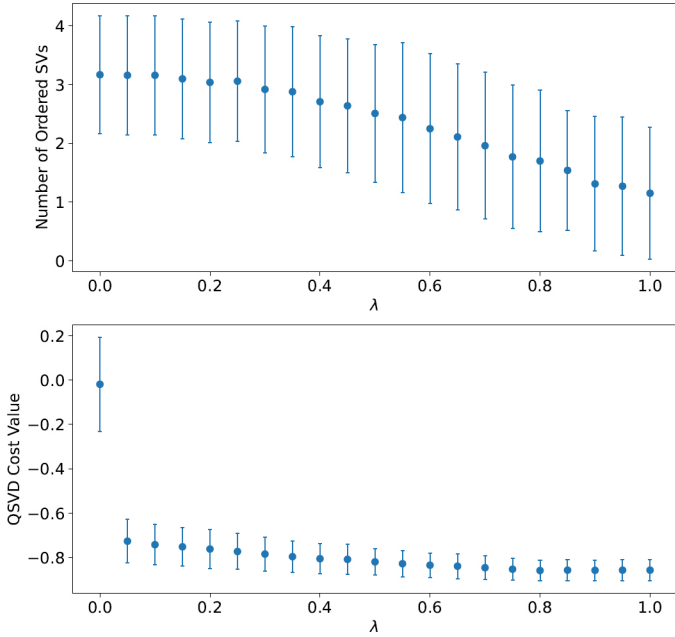


Fig. 4. Ordered QSVD Results for Random 4-qubit States. Top: Relationship between λ and the number of ordered singular values. The ordering property seemed to diminish at $\lambda \geq 0.15$. Bottom: Relationship between λ and the QSVD term value. All error bars show standard deviation.

B. PCA of Iris Dataset

One important application of ordered SVD is PCA. The principal components of a dataset can be calculated by finding the right singular vectors of the data matrix with zero column mean. Since PCA is usually used for feature extraction or dimension reduction, it is useful to know the principal components in order of the singular values.

The Iris Dataset is made up of 150 data points and each data point has 4 positive elements. 128 out of 150 data points were chosen and used to make a 128×4 matrix. The row vector of column vector means was subtracted from each row, creating the data matrix X . X was normalized and amplitude encoded.

The matrix is 128×4 , so the A register is chosen to be the top 2 qubits and B register the bottom 7. The 2-qubit QCA was used for the A register PQC and a 7-qubit 20-layer Real Amplitude Ansatz was used for the B register PQC. This type of ansatz was chosen as it only creates states with real amplitude, and X only has real values. λ was set to 0.15. Optimization of the cost function was done using the SPSSA algorithm implemented in Qiskit. Qiskit was also used for the quantum circuit simulation.

The training optimized the $\lambda = 0.15$ cost function to value -0.8811. The QSVD term value and ordering term values at the optimized parameters were -0.9116 and -0.8757 respectively. The calculated singular values were: $\{\sqrt{P|i}\}_{i=0}^3 =$

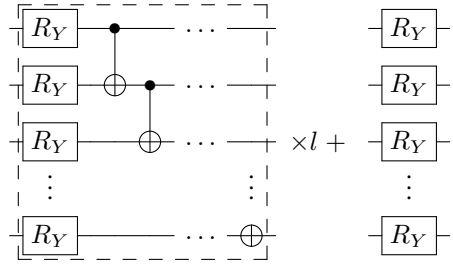


Fig. 5. The l -Layered Real Amplitude Ansatz. Each R_Y gate has a parameter, so for an n qubit l layered ansatz, there are $n(l + 1)$ parameters. This type of circuit was chosen since it can only create real amplitude states.

$\{0.9604, 0.1144, 0.1429, 0.2102\}$. Therefore the ordering was only successful up to the first singular vector.

The results are analyzed using the modulus of the inner products of the classically computed principal component with the i^{th} largest singular value ($\vec{\phi}_{c,i}$) and the principal component calculated using the quantum algorithm ($PQC_A^\dagger |j\rangle \rightarrow \vec{\phi}_{q,i}$). The inner product is used as a measure of how close the two vectors are. The modulus of this value is used because the quantum algorithm is expected to find the singular vectors up to a phase term. The principal components are thought to be found if these values are close to 1. The results are shown below. Since the singular vectors were not all ordered, the

TABLE I
MODULUS OF INNER PRODUCTS OF PRINCIPAL COMPONENTS

Choice of j	$i = 1$	$i = 2$	$i = 3$	$i = 4$
$j = i - 1$	1.0000	0.0519	0.5584	0.4078
i^{th} largest $P(j\rangle)$	1.0000	0.8753	0.5584	0.6138

choice $j = i - 1$ did not show values near 1 at all i values. However, when the correct ordering was chosen the values became close to 1 for $i = 1, 2$. Thus, the ordered QSVD found the first two principal components of the Iris data set, but did not order the second vector properly.

IV. DISCUSSION

A. Choosing the λ Value

The importance of knowing the singular vectors in order of their corresponding singular values comes from the fact that a fraction of them are relevant. Thus, we only need the relevant singular vectors to be ordered. Then depending on how many vectors are deemed to be relevant, the λ value can be chosen to have only the needed ordering while maintaining high SVD accuracy.

B. Customizing the Ordering Term

The difference of the ordered term value if the order of a pair of singular values are changed is written:

$$\begin{aligned} & |\{(\alpha_j P(|j\rangle) + \alpha_k P(|k\rangle))\} - \{\alpha_j P(|k\rangle) + \alpha_k P(|j\rangle)\}| \\ & = |(\alpha_j - \alpha_k) \{P(|j\rangle) - P(|k\rangle)\}| \end{aligned} \quad (11)$$

The cost function (10) has $|\alpha_j - \alpha_k| = 2|k - j|$. This is only dependent on the difference between the two indices. However, there may exist situations where this is disadvantageous. Since in general it is important to find the singular vectors with large corresponding singular values, we may want to design a cost function where the ordering of the larger singular values is rewarded more than the ordering of smaller singular values.

In this type of situation, multiple qubit expectation value terms may be used to help the ordering. Consider the following expectation value and measurement probability relation:

$$\langle Z_i \otimes Z_j \rangle = \sum_k (-1)^{b_i^k + b_j^k} P(|k\rangle) \quad (12)$$

Here b_i^k is the i^{th} least significant bit of the binary representation of k . We can consider a linear combination of the expectation values of two qubit observables.

$$\sum_{i,j} \gamma_{i,j} \langle Z_i \otimes Z_j \rangle = \sum_k \left(\sum_{i,j} \gamma_{i,j} (-1)^{b_i^k + b_j^k} \right) P(|k\rangle) \quad (13)$$

Compare the coefficient multiplied to each measurement probability with the (9) squared:

$$\sum_{i=1}^{n_A} \sum_{j=1}^{n_A} 2^{i+j-2} (-1)^{b_i^k + b_j^k} = \left\{ \left(\sum_{j=0}^{n_A-1} 2^j \right) - 2k \right\}^2 \quad (14)$$

With the choice $\gamma_{i,j} = 2^{i+j-2}$, the coefficient of $P(|k\rangle)$ becomes $\{(\sum_{j=0}^{n_A-1} 2^j) - 2k\}^2$ for (13). After adding this new term to the original ordering term, the difference of α in (11) becomes:

$$|\alpha_j - \alpha_k| = 4|j - k| \left| \frac{1}{2} - j - k + \sum_{i=0}^{n_A-1} 2^i \right| \quad (15)$$

This is dependent on $j + k$ and so we expect the singular vectors with higher singular values will be ordered better than ones with lower singular values.

This approach can be extended so that $\sum_i \alpha_i P(|i\rangle)$ has α_i be a polynomial function of i with highest order n_A . So, specific cost functions may be devised for specific ordering tasks. However, as the order of i is increased, the cost function becomes a global cost function. There is therefore a limit to the maximum order of the function $\alpha_i = f(i)$ to keep the quantum circuit trainable.

C. The Ordered QSVD as a Quantum Encoder

If OQSVD is successful, through training $PQCA, PQC_B$ becomes the transforms in (4) where $|u_i\rangle, |v_i\rangle$ are the singular vectors with the $(i+1)^{\text{th}}$ largest singular value. Thus, at the end of the optimized circuit, the input state $|\psi\rangle_{AB}$ becomes $\sum_i \sigma_i |i\rangle_A |0\rangle_B$. Here σ_i is the $(i+1)^{\text{th}}$ largest singular value. If we trace over the B register, we get the state $\sum_i \sigma_i |i\rangle_A$, and the original state in the A,B register has been encoded into a state in the A register. This has been discussed in [6].

OQSVD can take this process further. There are often cases where there are only few singular vectors with large, relevant singular values. If we take 2^m singular values into

account, then we can trace over some of the registers to get an approximated encoded state. Specifically,

$$\rho_{approx} = |0\rangle \langle 0|_{q_{A_1} \dots q_{A_m}} \otimes Tr_{q_{A_{m+1}} \dots q_{A_{n_A}}} [\rho_{QSVD}] \quad (16)$$

Here ρ_{QSVD} is the density matrix of $\sum_i \sigma_i |u_i\rangle_A$. For example, in the case of a 2-qubit ρ_{QSVD} and $m = 1$, the Frobenius distance between ρ_{QSVD} and ρ_{approx} is $\sqrt{2(|\sigma_2|^2 + |\sigma_3|^2)}$. This value should be small, since only two singular values are relevant. Thus, ρ_{approx} should be a good approximation of ρ_{QSVD} . The original quantum state can be encoded into a much smaller qubit state through this process.

V. CONCLUSION

We have suggested changes to the QSVD algorithm so that the singular vectors are known in order of their corresponding singular values after training. $\min(n_A, n_B)$ CNOT gates are added so that the QSVD term is calculated from the measurement of one of the registers. An additional ordering cost term uses the rearrangement inequality to force the wanted ordering. Constraints to keep the ordering term local were also discussed. Numerical simulations on random 4-qubit states showed that appropriate λ values could be chosen for the number of relevant singular vectors.

The proposed changes can be adopted for applications of QSVD where the larger singular values are important. Some examples may include PCA, feature extraction, dimension reduction, and removing noise from images. To illustrate this point, a numerical simulation of PCA on the Iris dataset was also shown.

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